

SIMATS ENGINEERING

SWETHA.B

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EXPERIMENT 10

ITA0606

MACHINE LEARNING

Write a Python Program to Implement Expectation & Maximization Algorithm.

Code

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture

# Generate sample data
np.random.seed(42)

X1 = np.random.normal(5, 1, (50, 2))
X2 = np.random.normal(10, 1, (50, 2))

X = np.vstack((X1, X2))

# Apply Expectation-Maximization algorithm
model = GaussianMixture(n_components=2, random_state=42)
model.fit(X)

# Predict clusters
labels = model.predict(X)

# Display cluster means
print("Cluster Means:")
print(model.means_)

# Display cluster assignments
print("\nCluster Labels:")
print(labels)

# Plot clusters
plt.scatter(X[:, 0], X[:, 1], c=labels)
plt.scatter(model.means_[0, 0], model.means_[0, 1],
            marker='x', s=200)
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("Expectation-Maximization Algorithm")
plt.show()
```

